Microeconomics I · Part I

Master in Economics · Universitat de Barcelona · 2026/27 · **Practice exam**

Name and surname

Rules

- **Part I exam: Wednesday 28 October 2026, 09:30–11:20, Room 1030.** 100 points: Part A 40, Part B 30, Part C 30.
- **This practice paper has the same length, format and rules as the real exam.**
- **Closed book.** You may use **pen, paper, any handwritten notes** (as many as you like, also photocopied or printed) and a **simple calculator**: basic arithmetic only, not programmable, not graphing, not a phone or any other connected device. You will not need one: every number works out by hand.
- **No typed or printed text** (typed notes, slides, books). **No formula sheet** is provided. **Phones and all other electronics** are switched off and put away.
- Keep your **photo ID** on the desk; it is checked at the start.
- Any answer that is **not supported by your calculation** receives only partial points or none (this does not apply to the multiple-choice questions).
- If you use additional sheets for calculations, attach them at the end and mark clearly which question each calculation belongs to.
- Do not separate the pages. Write your name on this page only.

Answering

- **Part A** (multiple choice): circle *one* letter in the answer box next to each question. Exactly one option is correct; a wrong answer costs nothing. To change an answer, cross out that circle and circle the new letter.

  Answer B: (A) (B) (C) (D) Changed from B to C: (A) ~~(B)~~ (C) (D)

- **Part B** (numbers): write the final number in the box and your working in the space below it.
- **Part C** (steps): write your steps in the boxes and the results in the answer boxes; each box is marked.
- **Numbers** (Parts B and C): an integer or an exact fraction (such as 7/3, in any equivalent form) is always accepted as the answer; you never need a decimal.
- Points are printed next to every question: about one point per minute. Notation: page 2.

DO NOT open before start

Notation and conventions

Notation as in class and in Mas-Colell, Whinston and Green (MWG): bundles $x = (x_1, x_2) \in \mathbb{R}^2_+$, prices $p = (p_1, p_2) \gg 0$, wealth $w > 0$; $x(p, w)$ Walrasian demand, $v(p, w)$ indirect utility, $h(p, u)$ Hicksian demand, $e(p, u)$ expenditure function.

For a price change from p^0 to p^1 with $u^0 = v(p^0, w)$ and $u^1 = v(p^1, w)$: compensating variation $CV = w - e(p^1, u^0)$ and equivalent variation $EV = e(p^0, u^1) - w$, both negative for a loss.

A lottery lists the probabilities of its prizes in the order given. Money is in euros.

Numbers (Part B, and the numbers in Part C): an integer or an exact fraction, in any equivalent form; you never need a decimal.

Extra space: write which question your working belongs to

Part A · multiple choice, 40 points · circle one letter in the answer box

A1 [4 points]

Bundles $x = (x_1, x_2)$ and $y = (y_1, y_2)$ lie in $\mathbb{R}^2_+$. Tom finds x at least as good as y if and only if $2x_1 + x_2 \geq y_1 + 2y_2$. Which statement about Tom's preferences is true?

A Complete: no. Transitive: yes.

B Complete: yes. Transitive: yes.

C Complete: no. Transitive: no.

D Complete: yes. Transitive: no.

Answer A1

A B C D

A2 [3 points]

Preferences on $\mathbb{R}^2_+$ are represented by $u(x_1, x_2) = \min\{2x_1 + x_2,\, 3x_2\}$. Which statement is true?

A They are convex, but not strictly convex.

B They are not convex: the indifference curves have kinks.

C They are strongly monotone.

D They are strictly convex.

Answer A2

A B C D

A3 [3 points]

Preferences on $\mathbb{R}^2_{++}$ are represented by $u(x_1, x_2) = x_1^{1/2} x_2^{3/2}$. The marginal rate of substitution is $\mathrm{MRS}_{12} = (\partial u/\partial x_1)/(\partial u/\partial x_2)$. What is MRS_{12} at $x = (2, 3)$?

A $\frac{1}{2}$

B $\frac{9}{2}$

C 2

D $\frac{3}{2}$

Answer A3

A B C D

Scratch for A3 (not marked)

A4 [3 points]

A utility function $u : \mathbb{R}^2_+ \to (0, 1)$ represents a consumer's preferences; every value of u lies strictly between 0 and 1. Which function also represents her preferences?

A $(u(x) - \frac{1}{2})^2$

B $\ln u(x)$

C $1 - u(x)$

D $u(x)\,(1 - u(x))$

Answer A4

A B C D

A5 [3 points]

A consumer has $u(x_1, x_2) = 4x_1 + 3x_2$, prices $p = (3, 2)$ and wealth $w = 18$. What does she buy?

A $(6,\, 0)$

B $(0,\, 9)$

C $(4,\, 3)$

D $(3,\, \frac{9}{2})$

Answer A5

A B C D

Scratch for A5 (not marked)

A6 [4 points]

Two goods. A consumer's preferences are locally non-satiated and her Walrasian demand $x(p, w)$ is differentiable. At prices $p = (9, 2)$ and wealth $w = 28$ she buys $x(p, w) = (2, 5)$. There, $\partial x_1/\partial p_1 = -1$ and $\partial x_1/\partial w = \frac{1}{12}$. What are $\partial x_2/\partial p_1$ and $\partial x_2/\partial w$ there?

A $\partial x_2/\partial p_1 = \frac{9}{2}$ and $\partial x_2/\partial w = \frac{1}{8}$

B $\partial x_2/\partial p_1 = \frac{9}{2}$ and $\partial x_2/\partial w = \frac{11}{12}$

C $\partial x_2/\partial p_1 = \frac{7}{2}$ and $\partial x_2/\partial w = \frac{11}{12}$

D $\partial x_2/\partial p_1 = \frac{7}{2}$ and $\partial x_2/\partial w = \frac{1}{8}$

Answer A6

A B C D

Scratch for A6 (not marked)

A7 [3 points]

Which property does every expenditure function $e(p, u)$ have?

A It is homogeneous of degree 1 in p.

B It is homogeneous of degree 0 in p.

C It is convex in p.

D It is decreasing in u.

Answer A7

A B C D

A8 [4 points]

Good 1 is normal for a consumer, with $\partial x_1/\partial w > 0$ at all prices and wealth. The price of good 1 rises; other prices and her wealth stay the same. Her compensating variation is $CV = w - e(p^1, u^0)$ and her equivalent variation is $EV = e(p^0, u^1) - w$. Which statement is true?

A $EV < CV < 0$

B $0 < CV < EV$

C $CV < EV < 0$

D $CV = EV < 0$

Answer A8

A B C D

A9 [3 points]

Zoe is an expected utility maximizer with an increasing, strictly concave Bernoulli utility u. A lottery F pays two different amounts with positive probability; $E[x]$ is its expected value and $c(F,u)$ its certainty equivalent. Which statement is true?

A $c(F,u) > E[x]$

B $u(c(F,u)) > E[u(x)]$

C $c(F,u) < E[x]$

D The risk premium $E[x] - c(F,u)$ is negative.

Answer A9

A B C D

A10 [3 points]

Prizes are 100, 50 and 0 euros. Ivan compares two lotteries first by the probability of 0 euros: the smaller, the better. Only if these probabilities are equal does he compare the probabilities of 100 euros: the larger, the better. Which axiom do his preferences violate?

A Transitivity.

B Continuity.

C Independence.

D Completeness.

Answer A10

A B C D

A11 [4 points]

Prizes are 240, 90 and 0 euros; a lottery lists their probabilities in this order. Eva is an expected utility maximizer with $u(240) = 1$ and $u(0) = 0$, and she is indifferent between 90 euros for sure and the lottery $(\frac{4}{5},\ 0,\ \frac{1}{5})$. Which lottery does she prefer to each of the other three?

A $(0,\ \frac{4}{5},\ \frac{1}{5})$

B $(\frac{3}{4},\ 0,\ \frac{1}{4})$

C $(\frac{1}{2},\ \frac{1}{5},\ \frac{3}{10})$

D $(\frac{1}{4},\ \frac{3}{4},\ 0)$

Scratch for A11 (not marked)

Answer A11

A B C D

A12 [3 points]

Lottery F pays 5 or 45 euros with probability $\frac{1}{2}$ each. Lottery G pays 5 euros with probability $\frac{1}{4}$, 25 euros with probability $\frac{1}{4}$ and 45 euros with probability $\frac{1}{2}$. Which statement is true? (CDF: cumulative distribution function.)

A Neither dominates: the CDFs cross.

B Neither dominates: the means are equal.

C F first-order stochastically dominates G.

D G first-order stochastically dominates F.

Answer A12

A B C D

Scratch for A12 (not marked)

Part B · numerical answers, 30 points · final number in the box, working below

B1 [4 points]

Preferences on $\mathbb{R}^2_+$ are represented by $u(x_1, x_2) = \sqrt{x_1} + 3\sqrt{x_2}$. Find x_1 such that the bundle $(x_1, 1)$ is indifferent to the bundle $(16, 4)$.

Answer B1: $x_1 =$

Your working (it must support the answer in the box)

B2 [4 points]

Preferences on $\mathbb{R}^2_+$ are represented by $u(x_1, x_2) = x_1^{1/3} + x_2^{1/3}$. The marginal rate of substitution is $\mathrm{MRS}_{12}(x) = \dfrac{\partial u/\partial x_1}{\partial u/\partial x_2}$. What is MRS_{12} at the bundle $x = (8, 27)$?

Answer B2: $\mathrm{MRS}_{12}(8, 27) =$

Your working (it must support the answer in the box)

B3 [4 points]

A consumer has utility $u(x_1, x_2) = 3\ln x_1 + 2\ln x_2$, prices $p = (3, 4)$ and wealth $w = 25$. How many units of good 1 does she buy?

Answer B3: $x_1 =$

Your working (it must support the answer in the box)

B4 [3 points]

A consumer's indirect utility function is $v(p, w) = \dfrac{w}{p_1 + 3p_2}$. What is her expenditure $e(p, u)$ at prices $p = (2, 1)$ and utility level $u = 6$?

Answer B4: $e =$

Your working (it must support the answer in the box)

B5 [4 points]

A consumer's expenditure function is $e(p,u) = 2u\,p_1^{3/4}p_2^{1/4}$. The price of good 1 rises from $p_1^0 = 1$ to $p_1^1 = 16$; $p_2 = 1$ stays (useful: $16^{1/4} = 2$). What is her true cost-of-living index $e(p^1,u^0)/e(p^0,u^0)$ (at any utility level u^0)?

Answer B5: index $=$

Your working (it must support the answer in the box)

B6 [4 points]

A consumer has utility $u(x_1, x_2) = \min\{3x_1, 4x_2\}$ and wealth $w = 63$. The price of good 2 is $p_2 = 8$; the price of good 1 rises from $p_1^0 = 3$ to $p_1^1 = 6$. What is her compensating variation CV of this price change (negative for a loss, as on page 2)?

Answer B6: $CV =$

Your working (it must support the answer in the box)

B7 [4 points]

Ivo is an expected utility maximizer with Bernoulli utility $u(x) = \ln x$ over money $x > 0$ in euros. A lottery pays 4 euros or 100 euros, each with probability $\frac{1}{2}$. What is its certainty equivalent, in euros? (You may use $\ln a + \ln b = \ln(ab)$.)

Answer B7: certainty equivalent $=$

Your working (it must support the answer in the box)

B8 [3 points]

Ivo (Bernoulli utility $u(x) = \ln x$) faces another lottery: 1 euro or 49 euros, each with probability $\frac{1}{2}$. What is its risk premium RP (expected value minus certainty equivalent), in euros?

Answer B8: $RP =$

Your working (it must support the answer in the box)

Problem C1 [16 points]

A consumer has the utility function $u(x_1, x_2) = x_1 + 6\sqrt{x_2}$ on $\mathbb{R}^2_+$. Write your steps in the boxes and the results in the answer boxes.

(a) [5 points] Find the Walrasian demand $x(p, w) = (x_1(p, w), x_2(p, w))$ for $w > 9p_1^2/p_2$.

steps for (a)

$x_1(p, w) =$ $x_2(p, w) =$

(b) [3 points] Find the indirect utility function $v(p, w)$ for $w > 9p_1^2/p_2$.

steps for (b)

$v(p, w) =$

(c) [4 points] Find the expenditure function $e(p, u)$ for $u > 18p_1/p_2$, and the Hicksian demands $h_1(p, u)$ and $h_2(p, u)$.

steps for (c)

$e(p, u) =$

$h_1(p, u) =$ $h_2(p, u) =$

(d) [4 points] Let $p_1 = 1$ and $w = 30$, and let the price of good 2 rise from $p_2^0 = 1$ to $p_2^1 = 3$. Compute her compensating variation CV and her equivalent variation EV of this price change.

steps for (d)

$CV =$ $EV =$

Problem C2 [14 points]

A consumer's preferences on $\mathbb{R}^2_+$ are represented by $u(x_1, x_2) = \max\{x_1,\, 2x_2\}$. Write your answers in the boxes.

(a) [3 points] Are these preferences monotone? Are they strongly monotone? Prove each answer, or give two bundles that show it.

monotone? strongly monotone? reasons or bundles for (a)

(b) [4 points] Are these preferences homothetic? Are they continuous? Prove each answer.

homothetic? continuous? reasons for (b)

(c) [3 points] Are these preferences convex? Circle Yes or No. Explain, or give bundles that show it.

Circle: Yes No

reason or bundles for (c)

(d) [4 points] Is there any concave utility function that represents these preferences? Circle Yes or No and explain.

Circle: Yes No

reason for (d)

Extra space: write which question your working belongs to. If you use additional sheets for calculations, attach them at the end and mark clearly which question each calculation belongs to.